

Rapid Communication

Aerobic expression of *Vitreoscilla* hemoglobin improves the growth performance of CHO-K1 cells

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Inefficient carbon metabolism is a relevant issue during the culture of mammalian cells for the production of biopharmaceuticals. Therefore, cell engineering strategies to improve the metabolic and growth performance of cell lines are needed. The expression of *Vitreoscilla stercoraria* hemoglobin (VHb) has been shown to significantly reduce overflow metabolism and improve the aerobic growth of bacteria. However, the effects of VHb on mammalian cells have been rarely studied. Here, the impact of VHb on growth and lactate accumulation during CHO-K1 cell culture was investigated. For this purpose, CHO-K1 cells were transfected with plasmids carrying the *vgb* or *gfp* gene to express VHb or green fluorescence protein (GFP), respectively. VHb expression increased the specific growth rate and biomass yields on glucose and glutamine by 60%, and reduced the amount of lactate produced per cell by 40%, compared to the GFP-expression controls. Immunofluorescence microscopy showed that VHb is distributed in the cytoplasm and organelles, which support the hypothesis that VHb could serve as an oxygen carrier, enhancing aerobic respiration. These results are useful for the development of better producing cell lines for industrial applications.

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1 Introduction

The metabolism of CHO cells is characterized by high glucose and glutamine uptake as well as high lactate and ammonium excretion rates. The accumulation of such byproducts is undesirable for bioprocess applications. Aerobic lactate production results from high glycolytic flux

[1]. Therefore, the basic strategies to avoid lactate production are based on decreasing the rate at which glucose is internalized [1]. A different approach to improve the metabolic performance of cells is to enhance the metabolism by the introduction of additional enzyme activities. Enhancement of the metabolism has been reported using the hemoglobin of *Vitreoscilla stercoraria* (VHb). VHb has been widely used to improve the microaerobic performance of prokaryotic and eukaryotic microorganisms [2]. It has been demonstrated that VHb also improves aerobic growth and strongly decreases the overflow metabolism of *E. coli* [3]. VHb is a hemoglobin that delivers oxygen faster than other hemoglobins due to a high oxygen dissociation rate [2]. VHb delivers the oxygen to the respiratory chain, enhancing ATP generation and NADH consumption at low oxygen concentrations [4]. This process could increase the carbon flux through the tricarboxylic acid (TCA) cycle [3]. It has been reported that VHb expression in CHO cells increased the specific productivity of the tissue plasminogen activator (tPA) but reduced the attained cell density

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Abbreviations: CHO, Chinese hamster ovary; GFP, green fluorescence protein; μ , specific growth rate; PBS, phosphate-buffered saline; q_{gln} , specific glutamine uptake rate; q_{lac} , specific lactate excretion rate; q_s , specific glucose uptake rate; *vgb*, *Vitreoscilla* hemoglobin gene; VHb, *Vitreoscilla* hemoglobin; $Y_{x/s}$, yield of biomass on glucose; $Y_{x/\text{gln}}$, yield of biomass on glutamine; $Y_{\text{lac}/x}$, yield of lactate on biomass

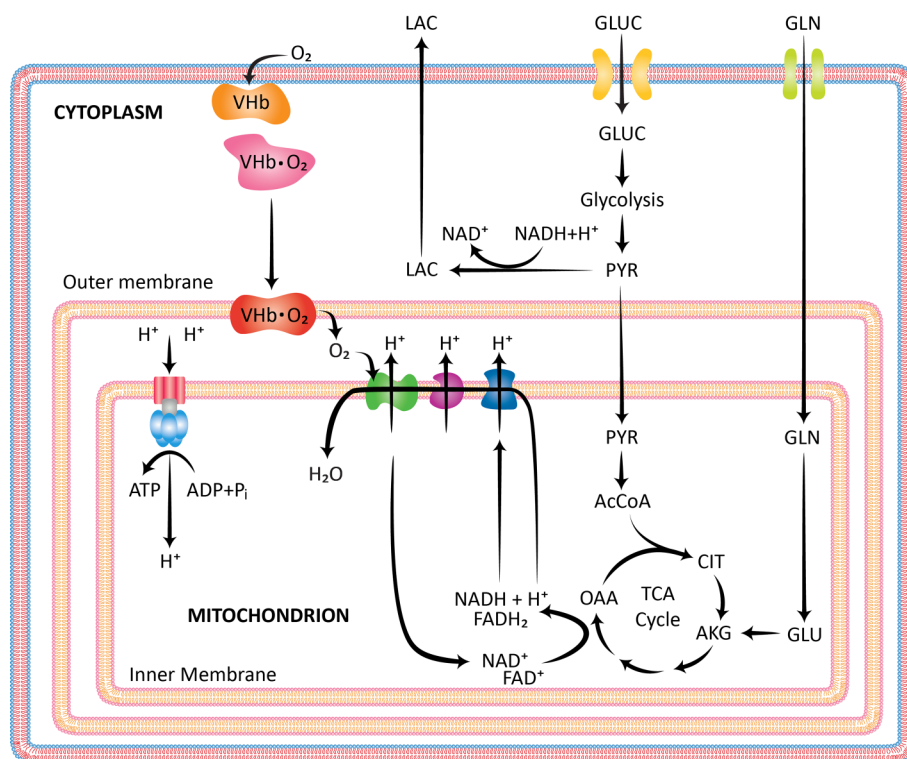


Figure 1. Possible role of VHb on the metabolism of CHO cells. VHb may bind to oxygen and deliver it to the cytochromes. This would enhance respiration and in consequence the regeneration of NAD⁺, activating the TCA cycle, and thus contributing to decrease lactate synthesis. AcCoA, acetyl-coenzyme A; ADP, adenosine diphosphate; AKG, α -ketoglutarate; ATP, adenosine triphosphate; CIT, citrate; GLN, glutamine; GLU, glutamate; GLUC, glucose; H₂O, water; LAC, lactate; NAD⁺, nicotinamide adenine dinucleotide (oxidized); NADH, nicotinamide adenine dinucleotide (reduced); OAA, oxaloacetate; Pi, inorganic phosphate; PYR, pyruvate; TCA, tricarboxylic acid. Adapted from [1].

[5]. Furthermore, the specific growth rate in VHb-expressing CHO cells was also reduced [5], possibly causing higher tPA production. Nevertheless, the authors did not provide more data on the use of carbon sources and lactate accumulation. In the present work, the potential of VHb to improve the aerobic performance of CHO cells was investigated. Because of the general effect of VHb on its host, a higher pyruvate flux to the TCA cycle may be expected, as well as lower lactate production (Fig. 1), which in turn could improve cell growth.

2 Materials and methods

2.1 Gene sequence optimization, synthesis and cloning

The VHb gene sequence (*vgb*, GenBank accession number: M30794) was optimized for expression in CHO cells using the Optimizer tool (<http://genomes.urv.es/OPTIMIZER/>). After codon optimization, the sequence GCCACC was included before the start codon to complete the Kozak sequence ACCAUGG [6]. Therefore, the second codon CTG (leucine) was changed to GTG (valine) because a guanine in the +4 position is important for the Kozak sequence [6]. The optimized sequence is provided in the Supporting information. The sequence was synthesized by GeneScript (NJ, USA) and subcloned into the pVAX1 vector (Thermo Fisher Scientific; CA, USA).

2.2 Plasmid amplification

Plasmids pVAX1-*vgb* and pVAX1-*gfp* were produced by an *Escherichia coli* W3110 *recA*⁻ strain grown in EnPresso B Plasmid growth system (BioSilta, Cambridgeshire, UK) following the instructions of the manufacturer. Plasmids were extracted using the plasmid MidiKit from Qiagen (Hilden, Germany) and quantified using the NanoDrop 2000 (Agilent Technologies, CA, USA). The plasmids supercoiled fraction, measured by agarose gel electrophoresis, was between 94 and 100%.

2.3 Cell culture and transfection procedure

CHO-K1 cells (ATCCR CCL-61TM) were cultured in F12-K medium (ATCCR 30-2004TM) supplemented with 10% fetal bovine serum (Biowest, Nuaille, France) and passaged twice per week at a split ratio of 1:16. The cells were cultured at 37°C in a 5% CO₂ atmosphere in a humidified incubator. One day prior to transfection, 250 000 cells in 2 mL of growth medium per well were seeded in 6-well plates, such that they were 70% confluent at the time of transfection. Different procedures for transfection were evaluated and reported in the Supporting Information file. The transfection protocol used was carried out with 2.5 μ g of pDNA and 6 μ L of Lipofectamine 2000 (Thermo Scientific, CA, USA) according to the manufacturer's instructions. For cell growth curve construction, four six-well plates were prepared for cell counting and media

sampling at 12, 24, 36 and 48 h post-transfection. The medium was changed 24 h post-transfection. The complete procedure was performed in triplicate using independently prepared cultures ($n = 3$).

2.4 Metabolite quantification and viable cell counting

Cells were counted in duplicate using the improved Neubauer hemocytometer and Trypan blue for viability measurement. Glucose, lactate and glutamine in the media samples were measured using an YSI 2700 biochemistry analyzer (YSI Inc.; OH, USA). Yields and rates were calculated by the proper mass balances over the time periods involved (12–48 h post-transfection).

2.5 Western blot

Samples were collected 24 h after transfection. Whole-cell protein extracts separated by SDS-PAGE were transferred to a nitrocellulose membrane (pore size of 0.2 μm). VHb was detected using a monoclonal antibody against VHb from mouse ascitic fluid (Abmart; Shanghai, China), which recognizes the amino acid sequence RPLFDMGROESL in VHb. The membrane was incubated overnight at 4°C with the anti-VHb antibody solution (0.15 ng/ μL). Next, the membrane was incubated for 1 h with the secondary antibody goat anti-mouse IgG (H+L) horseradish peroxidase conjugate (Thermo Fisher Scientific; OR, USA). Immunodetection was performed using a chemiluminescent substrate system (SuperSignal West Femto Maximum Sensitivity Substrate from Thermo Fisher Scientific; IL, USA). The protein extract from cells transfected with pVAX1-*gfp* was used as a negative control. Additionally, protein extract from *E. coli* BL21 expressing VHb and the antibody anti-actin (sc-1616; Santa Cruz Biotechnology; TX, USA) were used as positive and loading controls, respectively.

2.6 Immunofluorescence

CHO cells were grown on ethanol-washed coverslips, and the cells were transfected as described previously. 24 h after transfection, cells were fixed with 4% paraformaldehyde in PBS, and permeabilized in 0.1% TritonTM X-100/PBS. Next, Image-iTTM FX Signal Enhancer (Molecular Probes; OR, USA) was added to cover each coverslip. The cells were blocked with 1% BSA/PBS and then incubated with mouse anti-VHb antibody (Abmart; Shanghai, China, dilution 1:1200) in 1% BSA/PBS overnight at 4°C. The coverslips were washed and incubated for 1 h at room temperature with FITC-labeled goat anti-mouse antibodies (Jackson ImmunoResearch; PA, USA) in 1% BSA/PBS. Finally, the cells were stained with rhodamine phalloidin (Molecular Probes; OR, USA) to label F-actin according to the manufacturer's instructions. The preparations were

further washed with PBS and mounted on 10 μL of Vectashield[®] with DAPI (Vector Laboratories; CA, USA). Images were acquired using an Axiovert 40 CFL fluorescence microscope (Carl Zeiss; Göttingen, Germany).

3 Results

The original *vgb* sequence had only 30% of the codons in the highest frequency usage group (91–100%) for CHO cells. Moreover, the codon adaptation index (CAI) was 0.62, which is considered low because values nearest to 1 are correlated with higher expression levels [7]. The rare codon analysis indicated high probabilities for translation problems. After *vgb* optimization, the value of CAI was increased to 0.89, 64% of codons were in the highest frequency usage group, and no codons had percentages of frequency codon usage below 60% (Fig. 2A). Additionally, the GC content curve for the modified sequence did not show any peak outside the range between 30 and 70%, which is considered the ideal percentage range (Fig. 2B). The expression of the *vgb* optimized sequence in CHO cells was verified by Western blotting (Fig. 2C and 2D). Furthermore, VHb was detected in whole cells using immunofluorescence after transfection with pVAX1-*vgb*, and it was found that this protein is located throughout the cytoplasm and membranes (Fig. 2E). Similar amounts of GFP and VHb were produced by the cells, as shown in the Supporting Information file.

During the first 12 h post-transfection, there was no difference in the cell count of CHO cells expressing VHb or GFP (Fig. 3A), while a higher percentage of viable cells were observed when cells expressed the VHb, compared with GFP (Fig. 3B). After this time, higher cell counts were reached by cells expressing VHb, and its specific growth rate (μ) was 57% greater compared with cells expressing GFP (Fig. 3F). At 48 h post-transfection, cells expressing the VHb reached 2.8×10^6 cells/well, whereas those expressing GFP reached only 2.1×10^6 cells/well, and similar viability was observed for both conditions (Fig. 3A and 3B). At this time, the wells were saturated with cells, and the growth could not be monitored reliably anymore. No phototoxicity was found as effect of GFP expression, as shown in the Supporting Information. By the end of the culture, the consumption of glucose and glutamine, as well as lactate production were similar when VHb was expressed, compared with cultures expressing GFP (Fig. 3C and 3D). At 48 h after transfection only, the cells expressing VHb reached a lactate accumulation 12% higher than those cells expressing GFP (Fig. 3E). Although the specific glucose (q_s , Fig. 3H) and glutamine uptake rates (not shown) were similar for cells expressing VHb or GFP, the yields of biomass on glucose or glutamine were approximately 60% greater in cells expressing VHb (Fig. 3G and 3I). Noticeably, the yield of lactate on biomass ($Y_{\text{lac}/x}$) was 40% lower in cells expressing VHb than in cells expressing GFP (Fig. 3J).

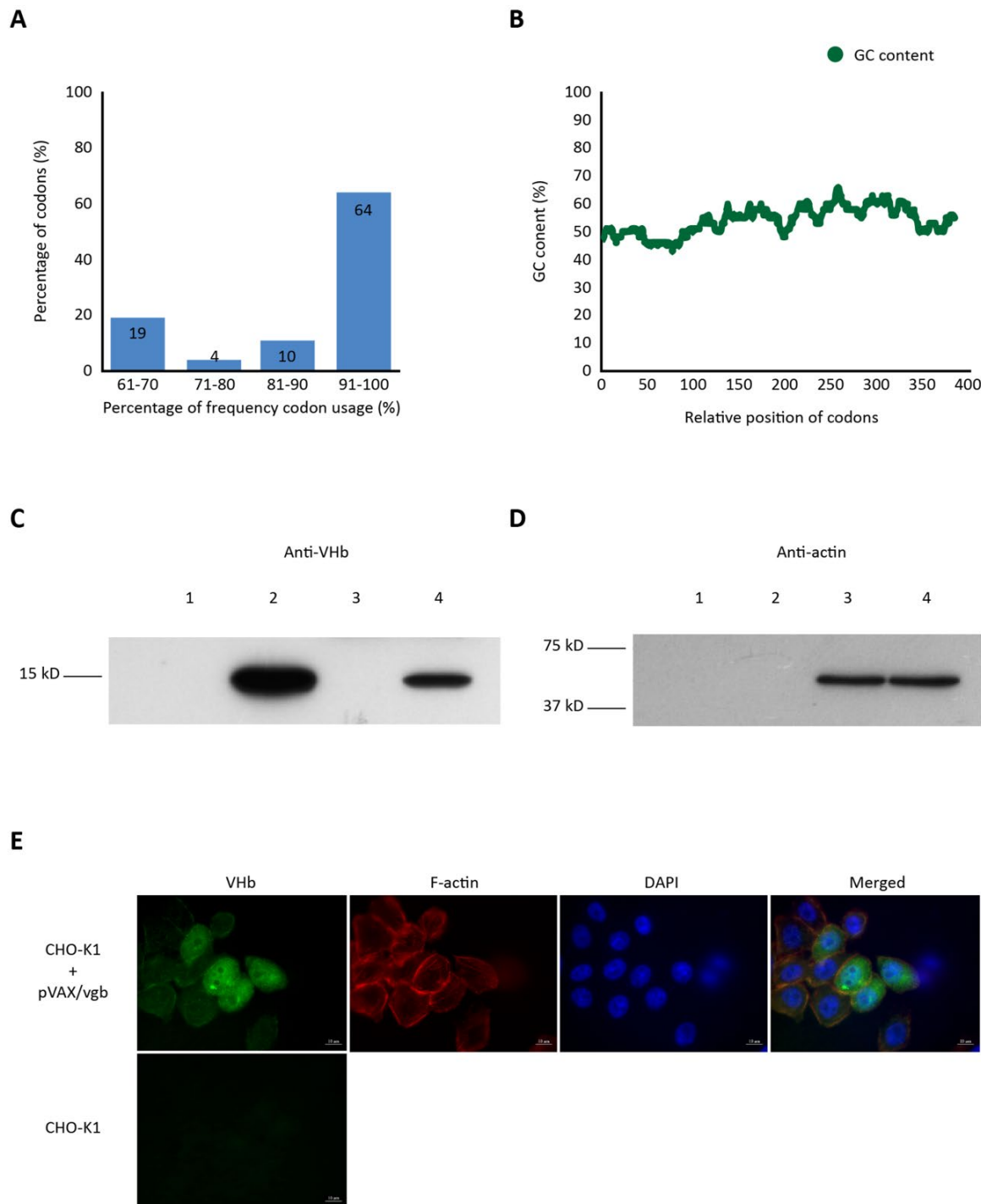


Figure 2. Sequence analysis of *vgb* gene after codon optimization and Western blot analysis of protein extracts. (A) Percentage distribution of codons arranged in groups with different usage frequency and (B) GC content distribution analyzed after codon optimization using the GeneScript (Piscataway, NJ) rare codon analysis tool. (C) and (D) Western blot of proteins evaluated by the anti-VHb and anti-actin antibodies. Lane 1: *E. coli* BL21 pBS; lane 2: *E. coli* BL21 pNKD1; lane 3: CHO-K1 pVAX1-*gfp*; lane 4: CHO-K1 pVAX1-*vgb*. The position of molecular size marker (kDa) is shown on the left. (E) Fluorescence microscopy of VHb from transfected CHO-K1 cells. Subcellular localization of VHb protein was detected in pVAX1-*vgb* transfected cells (upper panel) with the monoclonal anti-VHb antibody and a secondary FITC-labeled goat anti-mouse antibody (green) and is shown on the first column. The F-actin network was visualized in cells stained with rhodamine-labeled phalloidin (red) and is shown on the second column. The cells were stained with DAPI (blue) to visualize the nuclei and are shown in the third column, and the merged images are shown in the last column. VHb is distributed throughout the cytoplasm and in intracellular structures. Each micrograph is a representative image of $n = 3$ experiments. Negative control with CHO-K1 untransfected cells is shown at bottom panel. Scale bars 10 μ m.

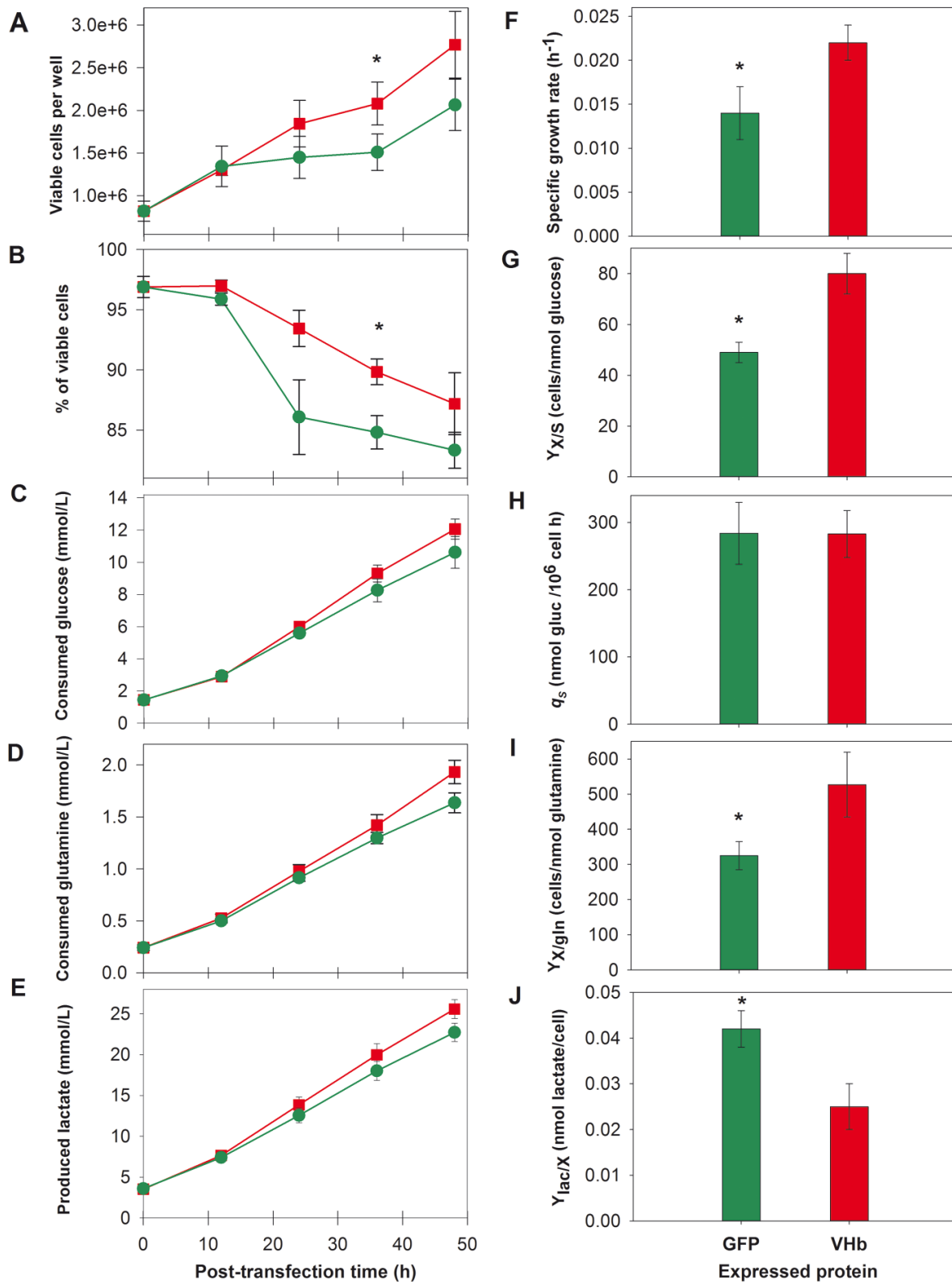


Figure 3. CHO-K1 cells culture profiles. (A) Cell growth curve of CHO-K1. (B) % of viable cells. (C) Consumed glucose. (D) Consumed glutamine. (E) Produced lactate. CHO cell expressing GFP or VHb are represented by circles or squares, respectively. (F) Specific growth rate. (G) Cells yield on glucose. (H) Specific glucose consumption rate. (I) Cells yield on glutamine. (J) Lactate yield on cells. Data are presented as the mean $\pm$ standard deviation ($n = 3$). $*p < 0.05$ was evaluated and significant difference confirmed.

4 Discussion

VHb was successfully expressed in CHO cells using an optimized synthetic *vgb* sequence. Interestingly, VHb was found to be distributed in the cytoplasm, which supports its proposed role as an intracellular oxygen facilitator. However, the technique used in the present work does not allow concluding whether VHb is also located inside the mitochondria, which is necessary for its proposed role in the respiratory chain. The diffusion of VHb in the cytoplasm is expected to be considerably lower than the diffusion of oxygen. In consequence, the abundance of VHb inside the cells should be much higher than intracellular concentration of oxygen to play an effective role as a facilitator of oxygen transport [8]. In lower eukaryotes such as yeast *Schwanniomyces occidentalis*, approximately 22.5% of the VHb was found in the mitochondria and 77.5% was found in the cytoplasm [9]. Therefore, it may be expected to find some VHb in the mitochondria of CHO cells.

Cells expressing VHb exhibited greater μ than cells expressing GFP, a finding that is coincident with reports about VHb improvement in cell growth in other eukaryotic organisms, such as *Pichia pastoris* [10, 11] and *Schwanniomyces occidentalis* [9], under aerobic conditions. However, the results of the present work contrast with the previous report of VHb expression in CHO cells, where no differences in growth were shown. Nonetheless, the previous study did not use an optimized *vgb* sequence and induced VHb expression using dexamethasone [5], making both studies not directly comparable. Glucose and glutamine were more efficiently utilized by cells expressing VHb than by cells expressing GFP, as evidenced by the biomass formation. Additionally, the amount of lactate produced per cell ($Y_{lac/X}$) was lower in cells expressing VHb than in those cells expressing GFP. Altogether, these results point to VHb expression being possibly useful to achieve high cell densities in CHO cell cultures. The increased μ also resulted in unchanged values of q_S , q_{gin} and q_{lac} when VHb was expressed compared with cultures expressing GFP. However, it should be noted that a transfection efficiency of only 37% was achieved. Therefore, the observed effects of VHb can be improved by increasing the fraction of the population expressing VHb and its intracellular concentration. The

results obtained in this work show that the aerobic expression of the VHb is a useful tool to improve the growth performance of CHO cells. This work represents a simple, yet efficient way to improve culture performance through cell engineering concepts in mammalian cells.

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The authors declare no financial or commercial conflict of interest.

5 References

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Cover illustration

The cover shows the in the one-pot conversion of free fatty acids and triglycerides, a lipase producing recombinant *Aspergillus oryzae* immobilized on matrix (sponge like matrix) can directly convert to bio-diesel from mixture of substrate. The cover is prepared by Jerome Amoah, Emmanuel Quayson, Shinji Hama, Ayumi Yoshida, Tomohisa Hasunuma, Chiaki Ogino and Akihiko Kondo authors of the article "Simultaneous conversion of free fatty acids and triglycerides to biodiesel by immobilized *Aspergillus oryzae* expressing *Fusarium heterosporum* lipase". (<http://dx.doi.org/10.1002/biot.201600400>).

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Review

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<https://doi.org/10.1002/biot.201600438>

Rapid Communication

Simultaneous conversion of free fatty acids and triglycerides to biodiesel by immobilized *Aspergillus oryzae* expressing *Fusarium heterosporum* lipase

Jerome Amoah, Emmanuel Quayson, Shinji Hama, Ayumi Yoshida, Tomohisa Hasunuma, Chiaki Ogino and Akihiko Kondo

<https://doi.org/10.1002/biot.201600400>